

HybridCrypto Project Overview

<https://github.com/elvistang199/HybridCrypto>

Introduction

HybridCrypto is a small console application that demonstrates **hybrid encryption**: data is encrypted with a symmetric cipher (AES) for efficiency, and the symmetric key is protected using a public-key algorithm (RSA). The project is written in C++ and depends on the [Crypto++ library](#) for all cryptographic primitives. HybridCrypto offers a simple menu-driven interface that lets users generate RSA keys, create and manage text files, encrypt and decrypt files, and view encrypted data.

Cryptographic methods

HybridCrypto uses two algorithms:

- **RSA (public-key encryption)**: The program generates a 2048-bit RSA key pair using Crypto++'s `AutoSeededRandomPool` and saves the private key in `private.key` and the public key in `public.key`. RSA is only used to encrypt and decrypt the AES key and initialization vector (IV). The `encryptFile()` function loads the public key and encrypts the randomly generated AES key using RSA with OAEP padding and SHA-256. During decryption the private key is loaded and used to recover the AES key.
- **AES in CBC mode (symmetric encryption)**: After generating an AES key and IV, the program encrypts the contents of a chosen `.txt` file using AES-128 in CBC mode. The AES key length and IV are stored in the output file after being encrypted with RSA. When decrypting, the program extracts the AES key and IV, decrypts them with RSA and then uses them to decrypt the ciphertext back into plaintext.

The program includes a `MAGIC_HEADER` constant to identify encrypted files and uses `std::filesystem` for cross-platform file handling.

Installing the Crypto++ header files

HybridCrypto requires the Crypto++ library to compile. The library provides the header files and static/dynamic libraries needed to build the program.

Windows (using vcpkg or Visual Studio)

Crypto++ can be installed via [vcpkg](#). The official installation instructions explain that you should first clone and bootstrap vcpkg, then install Crypto++:

1. Install Git if it is not already installed.
2. Open a Developer Command Prompt and run:

```
git clone https://github.com/Microsoft/vcpkg
cd vcpkg
bootstrap-vcpkg.bat
vcpkg integrate install
vcpkg install cryptopp
```

3. After installation, open the HybridCrypto.sln solution file in Visual Studio. Add the vcpkg include and library directories (vcpkg can integrate automatically). Build the solution; Visual Studio will link Crypto++ automatically. Alternatively, if you compile manually with `cl`, specify `-I` to the Crypto++ headers and link against `cryptopp.lib`.

macOS (Homebrew)

On macOS you can install Crypto++ using Homebrew. The Homebrew formula page shows that the package is named `cryptopp`; install it by running:

```
brew install cryptopp
```

This command installs the header files and libraries into Homebrew's prefix (usually `/usr/local` or `/opt/homebrew`). When compiling the program, supply the include and library paths, e.g.:

```
g++ hybrid_rsa_aes.cpp -o hybridcrypto -std=c++17 -I/usr/local/include
-L/usr/local/lib -lcryptopp
```

Linux (Ubuntu and other distributions)

Linux distributions provide Crypto++ in their package repositories. On Debian/Ubuntu you can install the development package with `apt-get`:

```
sudo apt-get update
sudo apt-get install libcryptopp-dev libcryptopp-doc libcryptopp-utils
```

This installs the header files, examples and a static library (`libcryptopp.a`). To compile on Linux:

```
g++ hybrid_rsa_aes.cpp -o hybridcrypto -std=c++17 -lcryptopp
```

If packages aren't available or you want the latest version, you can build from source. The Crypto++ installation guide explains that you can clone the repository and build using `make`:

```
git clone https://github.com/weidai11/cryptopp.git
cd cryptopp
make
sudo make install
```

After installation, compile the program as shown above.

Program usage and menu options

When you run the compiled program, it displays a menu. Users interact by entering numbers corresponding to tasks:

- **1 – Generate keys:** Generates a new 2048-bit RSA key pair and writes `private.key` and `public.key`.
- **2 – View a .txt file:** Lists available `.txt` files in the program's directory and displays the selected file's contents.
- **3 – Create a new .txt file:** Prompts for a filename and content, writes the content to disk.
- **4 – Delete a .txt file:** Lists `.txt` files and asks for confirmation before deleting.
- **5 – Encrypt a text file:** Requires that `public.key` exists. Lists `.txt` files, asks for the output encrypted filename, generates a random AES key and IV, encrypts the file with AES-CBC, encrypts the AES key/IV with RSA and writes the result to a `.enc` file.
- **6 – View encrypted file in raw bytes:** Displays the hexadecimal representation of a `.enc` file (for debugging).
- **7 – Decrypt an encrypted file:** Requires that `private.key` exists. Lists `.enc` files, prompts for an output filename, decrypts the AES key and IV with RSA, decrypts the ciphertext with AES-CBC and writes the plaintext to a `.txt` file.
- **0 – Quit:** Exit the program.

Compiling and running the program

After installing Crypto++ and preparing your environment:

1. **Obtain the source:** Clone the repository <https://github.com/elvistang199/HybridCrypto> or download the source file `hybrid_rsa_aes.cpp` and the Visual Studio project.
2. **Compile:** On Windows, open `HybridCrypto.sln` in Visual Studio and build the project (ensure `vcpkg` integration or manually set the include and library paths to Crypto++). On Linux or macOS, compile with `g++` using the appropriate include/library directories and the C++17 standard:

```
g++ hybrid_rsa_aes.cpp -o hybridcrypto -std=c++17 -lcryptopp
```

3. **Run:** Execute the resulting program. For example:

```
./hybridcrypto
```

The menu will appear. Choose option **1** to generate keys before encrypting or decrypting. For encryption or decryption you will be prompted for filenames and whether to delete the original file. The program writes encrypted files with the extension `.enc` and uses the `MAGIC_HEADER` to identify them【620991978159178†L0-L19】.

This document provides an overview of the HybridCrypto project, explains its hybrid encryption approach, details each menu option, and gives step-by-step instructions for installing the necessary Crypto++ header files on Windows, macOS and Linux before compiling and running the program.